QUANSER
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Mechatronic Sensors Trainer

Fundamental Labs – Weather Sensor

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Caution

This equipment is designed to be used for educational and research purposes and is not intended for use by the public. The user is responsible for ensuring that the equipment will be used by technically qualified personnel only. Users are responsible for certifying any modifications or additions they make to the default configuration.

Weather Sensor Lab

Why Explore Weather Sensors?

Weather sensors that measure things like temperature, pressure, and humidity play a vital role in how we interact with and understand our environment. These sensors are in a lot of devices we use in our day to day, from smartphones and smart thermostats to weather stations and aerospace systems. By having access to these environmental conditions as electrical signals, they enable real-time monitoring, data collection, and automated decision-making. Learning about the type of information these sensors give us and how to process it is essential when designing systems that respond to the physical world, like optimizing climate control or predicting weather patterns.

Background

This lab is part of the Fundamentals labs for the Mechatronic Sensors Trainer. These labs will guide you through the sensors that are part of the board and the ones that are provided as external sensors so you can understand how to read from the sensors, what they read, when they are used, and you feel comfortable using them. This will also include highlighting datasheets to learn how to interpret them and know what to look for.

Prior to starting this lab, please review the following concept review (located in Documents/Quanser/4_concept_reviews/),

- 4_concept_reviews/Mechatronics/**020_weather_sensor**.

Getting Started

In this lab, you will use the **Sensors Trainer** to learn about the basics of **Weather Sensor**. The lab will guide you through this sensor, how to read and use it as well as using the datasheet to understand it better.

Before you begin this lab, ensure that the following criteria are met:

- Make sure the Sensors Trainer has been setup and tested. See the Quick Start Guide (Quanser/2_quick_start_guides/sensors_trainer) for details on this step.
- You are familiar with the basics of the programming language you are using for this lab.
- Have your development environment ready,
    - If using Python, we recommend using Visual Studio Code.